

Marketing-roi-analysis using Simple Linear Regression

Importing Libraries

```
In [1]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import statsmodels.api as sm
```

Loading dataset

```
In [2]: df = pd.read_csv('Marketing ROI.csv')
df.head()
```

```
Out[2]:
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181

Data Cleaning and Exploration

Viewing data shape

```
In [3]: df.shape
```

```
Out[3]: (4572, 4)
```

Viewing data info

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   TV               4562 non-null   float64
1   Radio            4568 non-null   float64
2   Social_Media     4566 non-null   float64
3   Sales            4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

Checking for missing values

```
In [5]: df.isna().sum()
```

```
Out[5]: TV                10
        Radio              4
        Social_Media      6
        Sales              6
        dtype: int64
```

Handling/Removing missing values

```
In [6]: df=df.dropna()
```

Summary of numeric data

```
In [7]: df.describe()
```

```
Out[7]:
```

	TV	Radio	Social_Media	Sales
count	4546.000000	4546.000000	4546.000000	4546.000000
mean	54.062912	18.157533	3.323473	192.413332
std	26.104942	9.663260	2.211254	93.019873
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.555355	1.530822	112.434612
50%	53.000000	17.859513	3.055565	188.963678
75%	77.000000	25.640603	4.804919	272.324236
max	100.000000	48.871161	13.981662	364.079751

Heatmap diagram

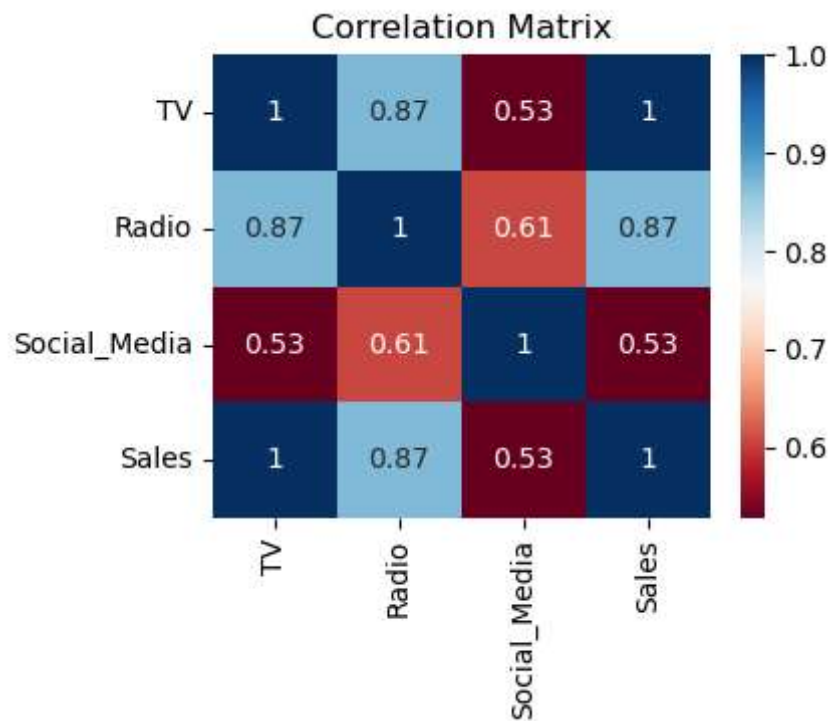
```
In [8]: plt.figure(figsize=(4,3))

sns.heatmap(df.corr(), annot=True, cmap='RdBu')

plt.title("Correlation Matrix")
```

```
plt.show()
```

```
print("This helps identify which advertising channel has the strongest relationship
```

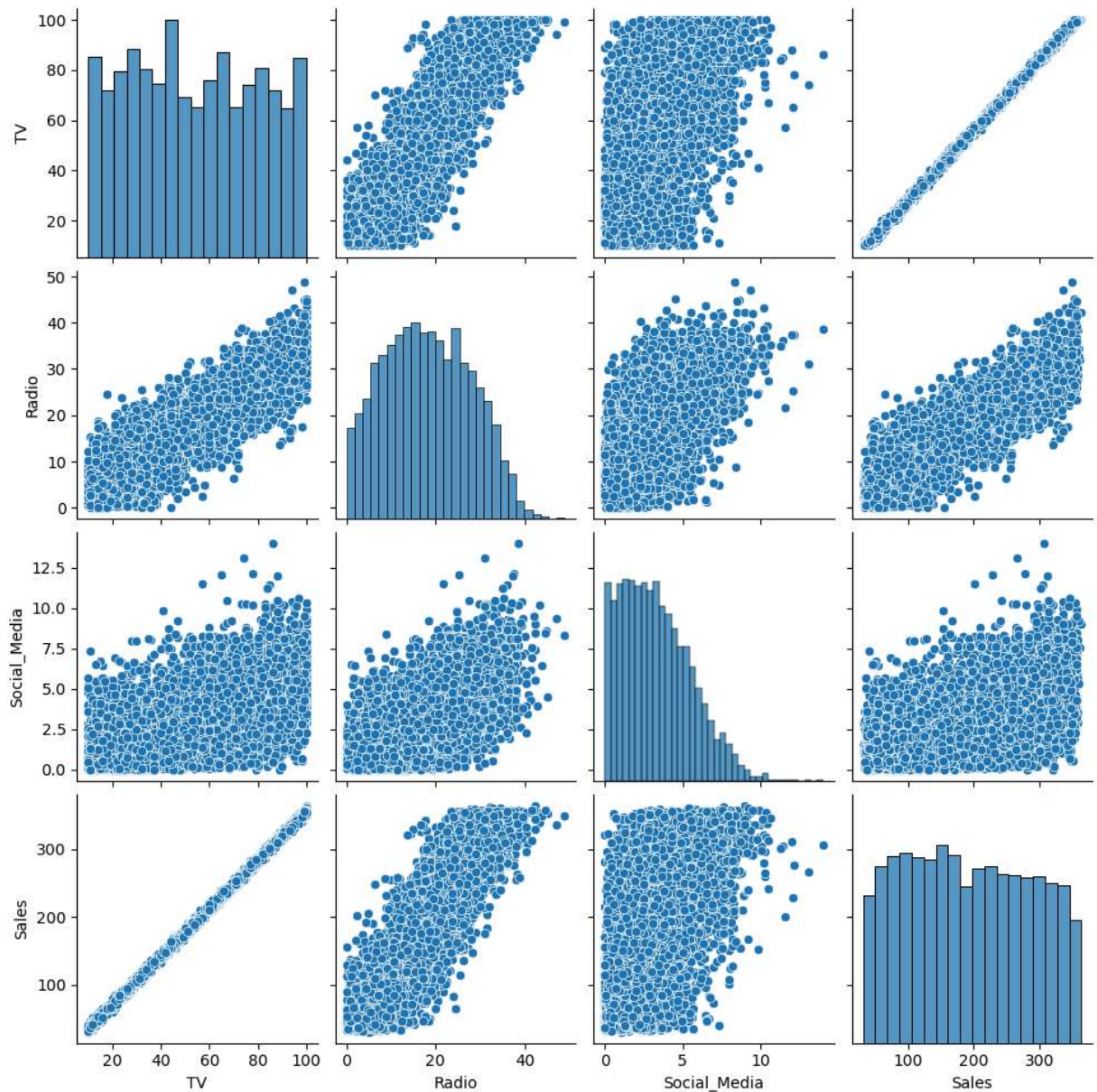


This helps identify which advertising channel has the strongest relationship with Sales.

Scatterplot to assess linear relationship between dataset

```
In [9]: sns.pairplot(df)
```

```
Out[9]: <seaborn.axisgrid.PairGrid at 0x22494986a50>
```



Correlation between variables

```
In [10]: print("Variable      Correlation with Sales")
print("-----")
print(" TV              1.00      ")
print(" Radio             0.87      ")
print(" Social Media       0.53      ")

print("TV is the best predictor because it has the highest positive correlation wit
```

Variable	Correlation with Sales
TV	1.00
Radio	0.87
Social Media	0.53

TV is the best predictor because it has the highest positive correlation with Sales.

Building the Simple Linear Regression Model

Using:

Independent Variable (X) → TV

Dependent Variable (y) → Sales

```
In [11]: X = df['TV']  
y = df['Sales']
```

Adding constant

```
In [12]: X = sm.add_constant(X)
```

Final formula

Sales = b(TV) + c

Fitting the model

```
In [13]: model = sm.OLS(y, X).fit()
```

Results

```
In [14]: model.summary()
```

Out[14]:

OLS Regression Results

Dep. Variable:	Sales	R-squared:	0.999			
Model:	OLS	Adj. R-squared:	0.999			
Method:	Least Squares	F-statistic:	4.517e+06			
Date:	Sat, 13 Jun 2026	Prob (F-statistic):	0.00			
Time:	15:58:45	Log-Likelihood:	-11366.			
No. Observations:	4546	AIC:	2.274e+04			
Df Residuals:	4544	BIC:	2.275e+04			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t 	[0.025	0.975]
const	-0.1325	0.101	-1.317	0.188	-0.330	0.065
TV	3.5615	0.002	2125.272	0.000	3.558	3.565
Omnibus:	0.052	Durbin-Watson:	1.998			
Prob(Omnibus):	0.974	Jarque-Bera (JB):	0.031			
Skew:	-0.001	Prob(JB):	0.985			
Kurtosis:	3.012	Cond. No.	138.			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Predicted Values

```
In [15]: predictions = model.predict(X)

predictions.head()
```

Out[15]: 0 56.851733
1 46.167191
2 145.889585
3 295.473177
4 53.290219
dtype: float64

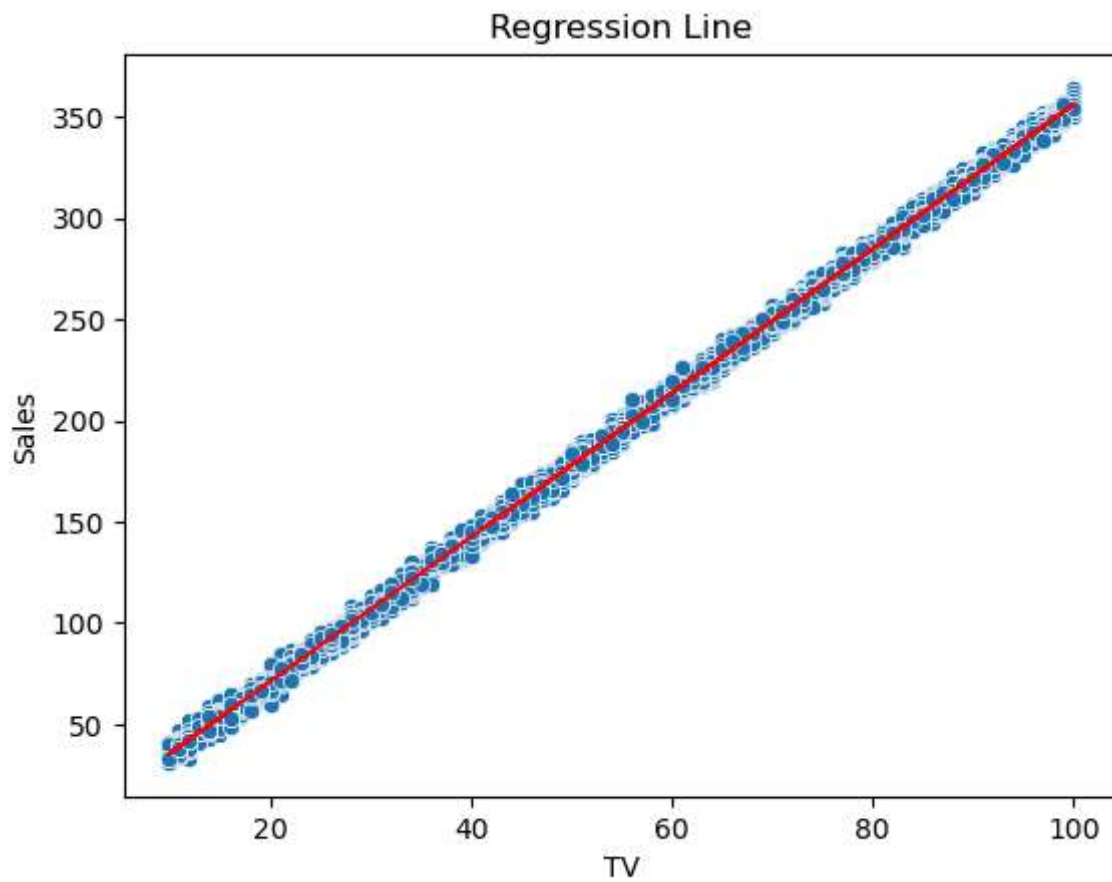
Diagnostic Plots

These help verify regression assumptions

Linearity Check

```
In [16]: sns.scatterplot(x=df['TV'], y=y)
plt.plot(df['TV'], predictions, color='red')

plt.title("Regression Line")
plt.show()
print("A straight-line trend indicates linearity")
```



A straight-line trend indicates linearity

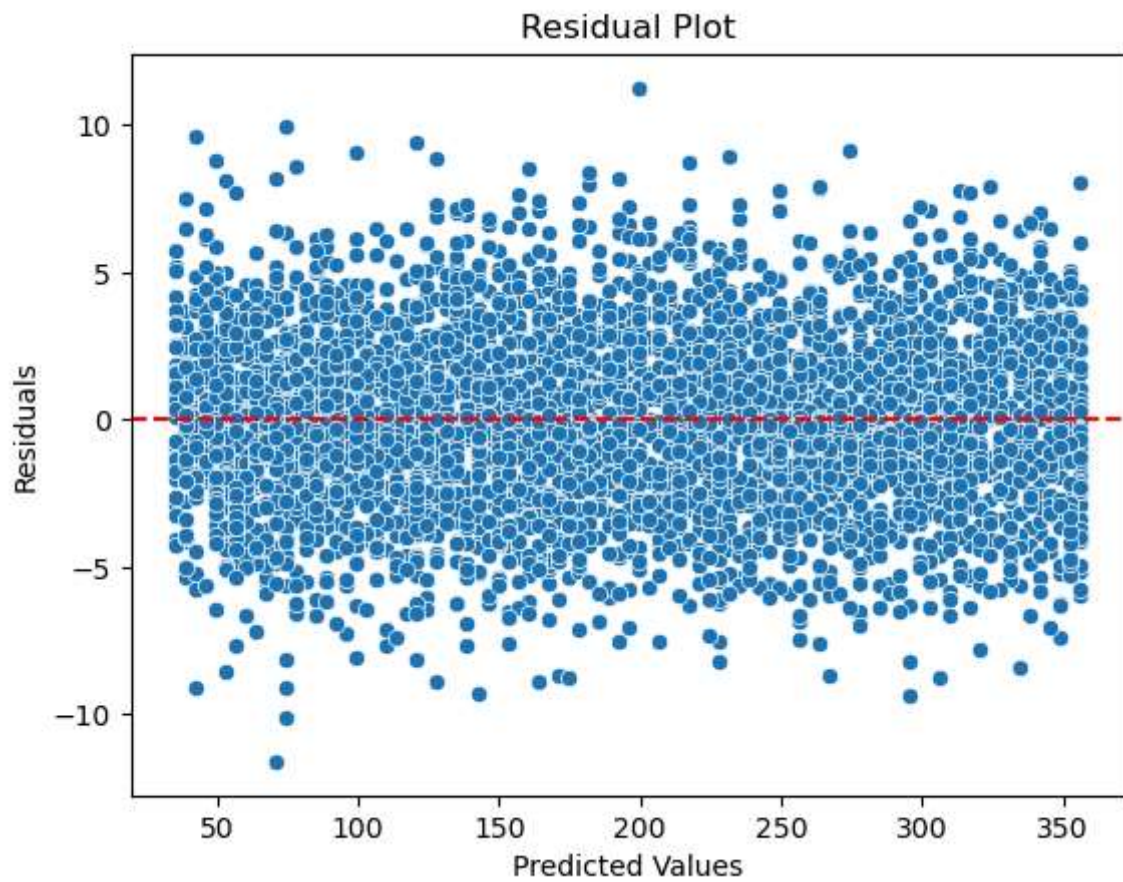
Residual Plot (Homoscedasticity)

```
In [17]: residuals = y - predictions

sns.scatterplot(x=predictions, y=residuals)

plt.axhline(y=0, color='red', linestyle='--')

plt.title("Residual Plot")
plt.xlabel("Predicted Values")
plt.ylabel("Residuals")
plt.show()
print("Residuals should appear randomly distributed around zero.")
```

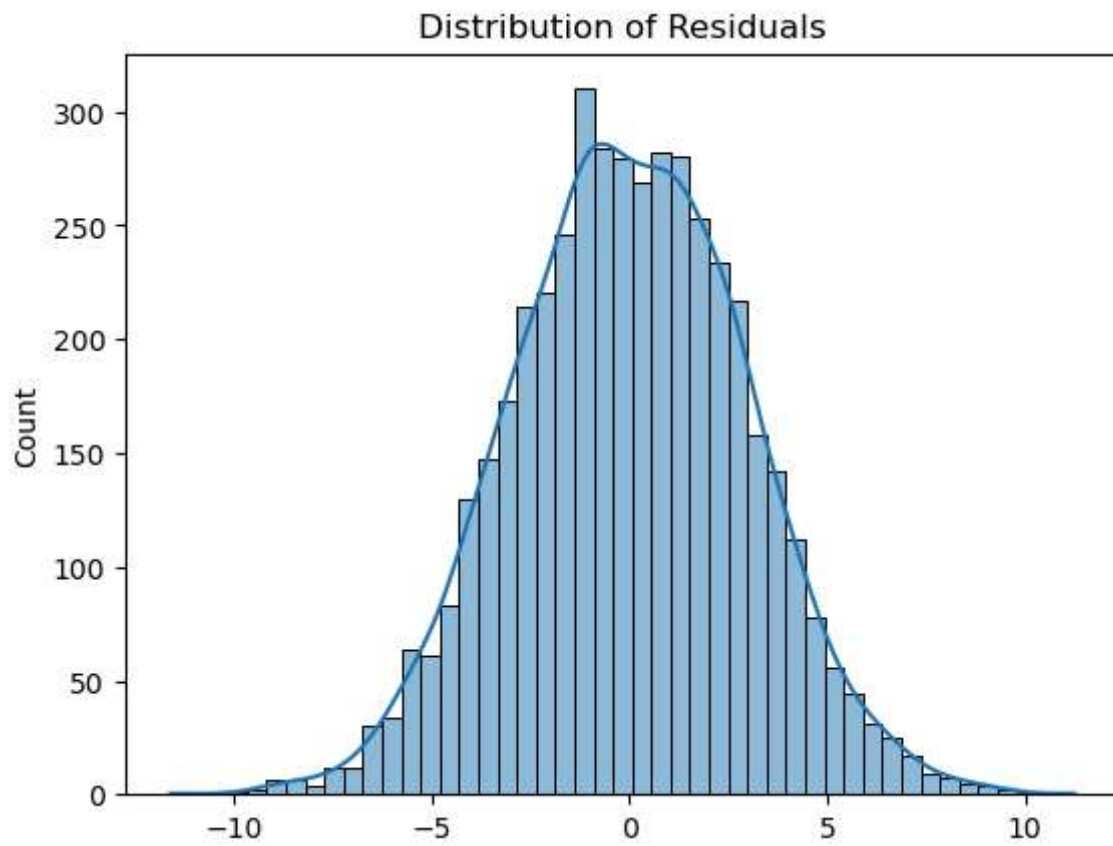



Residuals should appear randomly distributed around zero.

Normality Check

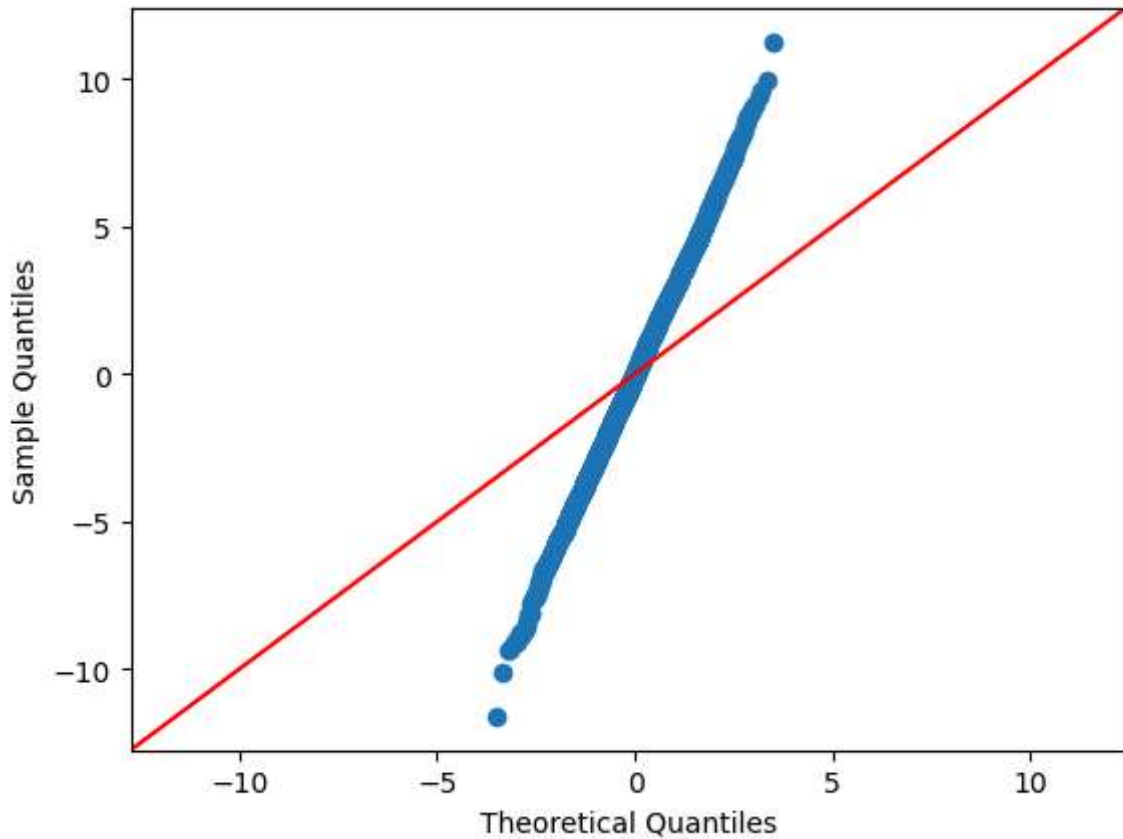
```
In [18]: sns.histplot(residuals, kde=True)

plt.title("Distribution of Residuals")
plt.show()
```

Q-Q Plot

```
In [19]: sm.qqplot(residuals, line='45')  
plt.show()
```



Recommended Marketing Strategy

Based on the regression analysis:

- TV advertising has the strongest relationship with Sales.
- The regression model is statistically significant.
- The high R-squared value indicates good predictive strength.
- Diagnostic plots confirm the model assumptions are reasonably satisfied.

ROI Recommendation

The company should allocate a larger percentage of its marketing budget toward TV advertising because it produces the strongest measurable impact on sales revenue.